

First-year Analysis Examination
Part Two
May 2022

Answer FOUR questions in detail.
State carefully any results used without proof.

1. Let $f : [a, b] \rightarrow \mathbb{R}$ have the property that for each $\varepsilon > 0$ there exist Riemann integrable functions ℓ and u on $[a, b]$ such that $\ell \leq f \leq u$ and $\int_a^b (u - \ell) < \varepsilon$. Does it follow that f is Riemann integrable on $[a, b]$? Proof or counterexample.
2. Suppose $(p_n)_{n=0}^\infty$ is a sequence of polynomials in one variable.
 - (i) Assume that $p_n \rightarrow f$ uniformly on $[0, 1]$ as $n \rightarrow \infty$; deduce as much as possible about the function $f : [0, 1] \rightarrow \mathbb{R}$.
 - (ii) Assume that $p_n \rightarrow f$ uniformly on $\mathbb{R}$ as $n \rightarrow \infty$; deduce as much as possible about the function $f : \mathbb{R} \rightarrow \mathbb{R}$.
3. Let the continuous function $f : [0, 1] \rightarrow \mathbb{R}$ satisfy

$$\int_0^1 f(t)t^n dt = 1/(n+2)$$

for all but finitely many positive integers n . Deduce as much as possible about f .

4. Let $(f_n)_{n=0}^\infty$ be a sequence of measurable real-valued functions on some measurable space. Prove that each of the following sets is measurable:
 - (i) $A = \{\omega : \sum_{n=0}^\infty f_n(\omega) \text{ is absolutely convergent}\}$;
 - (ii) $C = \{\omega : \sum_{n=0}^\infty f_n(\omega) \text{ is conditionally convergent}\}$.
 5. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space; let $A_0 \subseteq A_1 \subseteq \dots$ be an increasing sequence in $\mathcal{A}$ such that $\cup_{n=0}^\infty A_n = \Omega$; let $f : \Omega \rightarrow \mathbb{R}$ be measurable; and let L be a real number. Consider the statements:
 - (i) f is integrable on Ω and $\int_\Omega f d\mu = L$;
 - (ii) f is integrable on each A_n and $\int_{A_n} f d\mu \rightarrow L$ as $n \rightarrow \infty$.Prove that (i) implies (ii) and decide (with proof or counterexample) whether (ii) implies (i).
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